Exploring the Impact of State-Subsidized Childcare on Fertility Intentions and Family Planning in Germany

by

Anca Daniela Marineci

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Tilburg School of Economics and Management
Tilburg University

Supervised by: Dr Valentina Melentyeva

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Abstract

Global fertility rates have declined to historic lows, presenting severe macroeconomic vulnerabilities, particularly for Germany's labor supply and social security systems. A major driver of this decline is the "motherhood penalty," which frequently forces women to choose between career advancements and family formation. Consequently, many women alter their reproductive behavior by delaying childbirth to secure financial buffers. While extensive academic literature examines realized demographic outcomes, this thesis shifts the focus toward fertility intentions, examining the crucial link between family planning and childcare accessibility. In doing so, this study explores two distinct dimensions: correlation and causation.

To evaluate the correlational dynamics, a random-effects binary choice framework is applied to individual-level data. First, data from the German Socio-Economic Panel from 2004 to 2024 is analyzed to understand the relationship between parental satisfaction with the available childcare facilities and strict fertility timing. In addition, Waves 1 and 2 of the Generations and Gender Survey are used to evaluate how satisfaction with the domestic division of childcare tasks correlates with short-term fertility desires. Finally, to address causation, this paper implements a Synthetic Difference-in-Differences design, combining data from Germany's Regional Database with data from the German Socio-Economic Panel. Exploiting the 2008 Childcare Funding Act as an exogenous policy shock, we evaluate the causal effect of formal childcare expansion on fertility timing.

The regression models reveal that a one-unit increase in subjective childcare satisfaction is associated with a highly precise one percentage point increase in the probability of a planned pregnancy. Conversely, satisfaction with the division of domestic tasks yields a statistically insignificant relationship with short-term fertility desires, which remain heavily driven by baseline demographic traits such as age and education. The causal framework reveals a significant, consistent decrease in the rate of planned pregnancies following the infrastructure expansion.

Rather than demonstrating a reduced desire for children, these findings indicate that expanding formal childcare successfully acts as a structural buffer. Access to guaranteed institutional care absorbs domestic burdens and mitigates the career threats of an unplanned pregnancy, easing the stressful need for women to rigidly schedule their transition into motherhood. Ultimately, this thesis underscores that alleviating the motherhood penalty through investments in institutional infrastructure reduces the rigid constraints of family planning, empowering women to freely manage their transition into parenthood.

Technology Statement

During the preparation of this work, I used Google's Gemini 3.1 Pro Model in order to check my text for grammar and spelling, fix bugs in my code, and refine the structure of my paper. The content, interpretations, model description and econometric conclusions remain entirely my own work. The following parts of the assignment were affected by AI tool usage: All 7 Chapters of the Bachelor Thesis. After using this tool/service, Anca Daniela Marineci evaluated the validity of the tool's outputs, including the sources that generative AI tools have used, and edited the content as needed. As a consequence, Anca Daniela Marineci takes full responsibility for the content of their work.

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Chapter 1

Introduction

1.1 Problem Description

The worldwide community is currently experiencing an unprecedented demographic paradigm, characterized by global fertility rates that have now declined to historical lows. As individuals increasingly face hardships, uncertainties, and the demanding pace of contemporary life, there is a marked, growing reluctance to transition into parenthood. Consequently, the decision to delay or entirely give up having children can no longer be viewed as a mere statistical anomaly or a transient cultural phase. Acknowledging and confronting this shift in reproductive behavior is essential for understanding the future trajectory of our societal structures.

Germany is not far different from the general global situation. According to the Federal Statistical Office (Statistisches Bundesamt), the total fertility rate fell to 1.35 children per woman in 2025, which is well below the 2.1 required for replacement. This carries profound macroeconomic and structural implications for German society (Federal Statistical Office, 2025).

Germany's economic system has historically relied on a robust domestic labor force, yet this demographic decline threatens to create labor shortages. The working-age share of the population is continuously shrinking, shifting the old-age dependency ratio and creating extreme tightness in the labor market. As the baby-boomer generation retires, the influx of younger cohorts is mathematically insufficient to replace the exiting human capital (Börsch-Supan, 2003). Perhaps the most acute economic vulnerability associated with low fertility rates in Germany is the sustainability of its pension and social security systems. An aging population places immense financial strain on a state where workers directly fund the retirement benefits of pensioners (Sinn, 2000).

A key determinant driving this issue is the "motherhood penalty," which refers to the significant financial and career setbacks that women experience after childbirth. Labor market research indicates that mothers face substantial earnings losses compared to childless individuals. For instance, Budig and England (2001) identify a 7 percent wage penalty for every child a woman has, and this functions through several mechanisms. Women are pushed to interrupt their careers for motherhood, leading to a loss of valuable job experience and seniority. Unlike past generations, women today are more aware that motherhood frequently leads to these reduced working hours, drops in labor force participation, and employer discrimination. This severe career penalty makes women less likely to want children, pushing fertility rates to an all-time low as they are forced to choose between financial stability and having a family (Feldhoff, 2021; Budig and England, 2001).

1.2 Research Question and Contribution

Because fertility rates have an immense impact on the current economic structure of our society, there is a significant body of academic literature available on the topic. However, these studies primarily focus on realized demographic outcomes, such as the determinants of fertility rates, the impact of the motherhood penalty, and the persistence of traditional cultural norms regarding maternal employment. In this study, we aim to contribute to the available knowledge by shifting the focus toward a critical precursor of these outcomes: fertility intentions.

Ultimately, we are interested in answering the following **research question**: "What is the relationship between fertility intentions and childcare accessibility?". In order to study this broader topic, we first divide our analysis into two dimensions: **correlation** and **causation**.

To examine the correlational aspect, we employ two sub-questions: "To what extent is parental satisfaction with the available childcare facilities correlated with fertility timing?" and "To what extent is satisfaction with the division of childcare responsibilities between spouses correlated with short-term fertility desires?". We choose this approach because, as we will detail below, there are two distinct mechanisms at play when studying fertility intentions. The first is fertility desire, sometimes studied as the wish to have a child in the next three years, which serves as a direct precursor to the total fertility rate. The second is fertility timing, which is not always synchronized with the total fertility rate. This refers to the rigid planning women undergo, in which they strictly time their pregnancies to avoid financial losses such as the motherhood penalty. Finally, to address causation, we ask the question: "What is the causal effect of formal childcare expansion on fertility timing?". Due to data limitations, our causal analysis focuses exclusively on fertility timing.

Specifically, we seek to understand whether the availability of state-subsidized childcare reduces women's need to explicitly time maternity, which can be a stressful process fueled by the fear of the motherhood penalty. While access to childcare facilities may not entirely erase career disruptions, it acts as a vital structural buffer. By providing a legal right to a kindergarten or nursery spot, the state absorbs some of the domestic responsibilities, allowing women to maintain control over their careers even in the event of an unplanned or early pregnancy. Furthermore, we are interested in identifying the key drivers of fertility desires. In contexts where data on institutional childcare is missing, we examine how the division of domestic childcare tasks affects family planning.

1.3 Literature Review

As indicated above, the transition into motherhood often introduces a significant conflict between career advancement and family responsibilities. A robust body of literature highlights that mothers frequently face long-term professional and financial disadvantages in the labor market.

Aware of these professional conflicts, many women have altered their reproductive behavior, largely by postponing parenthood. Mills et al. (2011) identify that the conflict between career advancement and parenthood, as well as the wage penalty mothers face, determine women to delay having children. Similarly, Miller (2011) demonstrates that delaying motherhood provides a substantial financial benefit. Specifically, each year of delayed motherhood increases a woman's career earnings by 9 percent and her wage rates by 3 percent. This advantage is particularly strong for college-educated women and those in professional or managerial occupations. Sonfield et al. (2013) emphasize that unplanned pregnancies are directly tied to negative economic outcomes. Women who enter into motherhood early or unexpectedly face severe wage penalties, are less likely to achieve their professional goals and have higher chances of requiring social assistance.

Consequently, demographic research has increasingly focused on how family policies, particularly childcare availability, might ease the structural obstacles that force women to choose between advancing in their careers and raising a family. Initial empirical investigations successfully established a baseline correlation between childcare infrastructure and fertility intentions. In China, Wang et al. (2026) demonstrated that regional kindergarten expansion directly supports parents' career aspirations and significantly increases their desires to have children. The research identifies two primary mechanisms for this relationship. First, accessible, high-quality childcare supports parents' career aspirations, particularly reducing work-family conflict and second, it increases parents' access to leisure time, effectively reducing the stress associated with childcare.

Analyzing comprehensive policy reforms in Germany, Bauernschuster et al. (2013) found that significant childcare expansion directly generated higher birth rates, refuting the assumption that such policies merely shift the timing of births without changing overall family size. Specifically, the estimates suggest that a 10-percentage-point increase in public childcare coverage leads to an increase of 1.4 births per 1,000 women. This dynamic is echoed in Norway, where Rindfuss et al. (2010) proved that transitioning to widespread, affordable childcare caused significant increases in fertility rates across all parities. The models suggest that shifting from zero childcare availability to a scenario where 60 percent of preschool children have access to affordable, high-quality care increases the average number of children a woman has by 0.5 to 0.7 by age 35.

Furthermore, in regions where state childcare is fragmented or culturally underutilized, informal support networks, the division of domestic labor and workplace dynamics become the primary determinants of fertility. Yoon (2017) highlights that in South Korea, intensive daily childcare assistance from co-residing grandparents, coupled with substantial male participation in household chores, acts as a critical substitute for formal state support to drive second births. Thomas et al. (2023) find that for highly educated, working women, the division of housework and childcare tasks becomes a significant predictor of fertility. The data showed that the probability of a couple where the woman is university-educated and employed having another child within five years is 73 percent if they have an equal division of housework. An additional finding was that these significant effects were primarily observed when the wives were the ones surveyed, rather than the husbands. The authors suggest that this discrepancy likely occurs because husbands tend to overestimate their own contributions to household chores, making men's self-reported data on domestic labor less reliable for predicting actual family outcomes. Finally, autonomy in the workplace also acts as a substitute for formal care provision. Across Europe, Begall and Mills (2011) showed that mothers possessing high subjective control and autonomy over their work schedules are much more likely to plan a second child.

Despite these causal links, the literature reveals that the benefits of childcare and family policies are heavily stratified across different demographics. Scherer et al. (2023) underline that public childcare availability shows slight positive effects on first births for younger women and second births for women with mid-level education, while private childcare availability is associated with higher-order births among individuals with higher education. In South Korea, Choi et al. (2018) found that just the implementation of policies such as maternity leave or flexible schedules do not universally encourage higher fertility intentions. Specifically, a woman's age heavily impacts whether such a policy changes her behavior or not. For working women under the age of 35, the availability of childcare leave, family allowances, and workplace daycare facilities significantly elevates their intentions to have children. Conversely, for women aged 35 and older, these same policies fail to stimulate childbearing desires, with workplace daycare even showing a negative association.

Therefore, the literature establishes that motherhood introduces a profound conflict between career advancement and family responsibilities, resulting in significant financial penalties. To mitigate these disadvantages, women frequently delay parenthood to build a professional and financial buffer for the inevitable disruptions. Consequently, an unintended pregnancy breaks up this strategy, exposing women to economic vulnerability and accelerated wage penalties. While the expansion of state-supported childcare is proven to aid maternal employment and help women balance their careers, it remains unclear how this institutional support influences the underlying stress of family planning. This study addresses this gap by examining whether increased childcare availability reduces the strict necessity for meticulously planned pregnancies, ultimately equipping women to better manage the career implications of an unintended transition into motherhood. Moreover, when information about institutional childcare is missing, we study the relationship between household-level responsibilities and short-term fertility intentions.

Chapter 2

Institutional Background

This section provides a detailed overview of the setting that shapes family planning and childcare attitudes in Germany. To understand how childcare accessibility impacts fertility intentions today, we must first understand the legal framework of the 2008 Childcare Funding Act. Additionally, this chapter outlines how the legacy of Germany's East-West divide created lasting regional differences in both institutional availability and cultural norms surrounding working mothers.

2.1 The 2008 Childcare Funding Act

A policy that was designed to erase the structural barriers of family planning is the Kinderförderungsgesetz (commonly abbreviated as KiföG, translated as the "Childcare Funding Act"), implemented on December 10th, 2008 (Bundesrepublik Deutschland, 2008). KiföG was designed to fundamentally improve the reconciliation of work and family life, enhance early childhood development, and facilitate the faster re-entry of mothers into the labor market. Under this act, the legal rights to childcare are vastly expanded. For children under one, parents become entitled to a childcare spot (either in a facility or with a childminder) if the child needed it for their personal development, or if the parents were working, seeking work, or in an educational/training program. Crucially, for children aged between one and three, the law established a strict legal right to early childhood support in a daycare center or through a childminder entirely independent of their parents' employment status (Bundesrepublik Deutschland, 2008).

Although this legal right to a state-subsidized childcare spot did not officially take effect until August 2013, municipalities were immediately tasked with building the extensive infrastructure required to sustain it. Consequently, the childcare expansion driven by this policy is effectively measured from 2008. To help local municipalities afford this massive infrastructure project, Article 3 of the Act established a special federal fund called the "Bundessondervermögen Kinderbetreuungsausbau" (translated "Special Federal Fund for Childcare Expansion"). Through this, the federal government committed to providing up to 2.15 billion Euros between 2008 and 2013 to subsidize the creation of new facilities and childminding spots. (Bundesrepublik Deutschland, 2008).

2.2 The Legacy of Germany's East-West Divide

However, the implementation of the KiföG met vastly different reactions across German states due to deep-rooted historical divisions regarding female employment and family structures, which we will explore in detail below. When KiföG introduced a legal right to a childcare place, East German states viewed it as a welcomed support for an already established system. Western states, however, faced immense pressure, as they had to rapidly build infrastructure and shift deeply ingrained cultural perceptions about institutional childcare for very young children.

These diverging cultural practices are rooted in the aftermath of the Second World War, when Germany as we know it today was split into two territories, commonly referred to as East and West Germany. The Federal Republic of Germany (FRG), established on May 23rd, 1949, was a capitalist democracy, while the German Democratic Republic (GDR), established on October 7th 1949, was a Marxist-Leninist socialist republic. For four decades, these two independent states developed entirely opposing institutional frameworks regarding family organisation and women's labour market participation. On October 3rd 1990, the country was reunited, but the very different policies had already left a significant mark on the cultural practices, local mentalities and imposed gender roles of the two sides (Kreyenfeld and Konietzka, 2017).

The trajectory of women's labor market participation, fertility dynamics, and work culture in Germany presents one of the most complex natural experiments in modern demographic and economic history. We are interested to know if geographical differences still persist today and in order to investigate this, we must first understand the historical context and cultural baselines of the two sides.

When looking at the 16 states which currently form Germany, we identify that West Germany was composed of the following: Baden-Württemberg, Bayern, Bremen, Hamburg, Hessen, Niedersachsen, Nordrhein-Westfalen, Rheinland-Pfalz, Saarland, Schleswig-Holstein and West Berlin, which was not considered a state, and is now unified with East Berlin. The FRG was constructed as a prototypical conservative welfare state, and institutionalized the "male breadwinner/ female homemaker" socio-economic model. This was made explicit through the federal tax system, specifically the mechanism of joint taxation which heavily penalized the secondary earner in a married household (Geist, 2009; Kreyenfeld and Konietzka, 2017). Women were discouraged to return to work due to this financial impediment, but also through popularized social norms and linguistic markers of stigma. Mothers who returned to their workplace full time after having children were referred to as "Rabenmutter" (literally translating to "raven mother"), suggesting they were unnaturally abandoning the young to fend for themselves. Additionally, institutional daycare centers were dismissively and pejoratively referred to as "Fremdbetreuung" (translating to "care by strangers"). This terminology further enforced the idea that the biological mother was the only appropriate caregiver for infants and toddlers (Boelmann et al., 2021).

On the opposite side, we have East Germany, consisting of now: Brandenburg, Mecklenburg-Vorpommern, Sachsen, Sachsen-Anhalt, Thüringen and East Berlin. The socialist republic considered labour participation an individual right, and for this reason, the number of working women was extremely high, in parallel with motherhood being also a universal phenomenon. This attitude most likely came from an urgent macroeconomic necessity, as the state simply wanted to maximize their labor supply. In order to achieve this, the GDR absorbed part of women's domestic responsibilities, by heavily investing in state-subsidized childcare facilities (Geist, 2009). "Krippen" (translated "nurseries") were universally available and fundamentally designed to enable mothers to return to work almost immediately after giving birth (Kreyenfeld and Konietzka, 2017). Although society expected women to work full time in order to contribute to the state's production, the domestic expectations for women living in East Germany remained almost as high as those for women living in the West. East German women developed a "dual orientation", because intensive family life and full-time employment became the baseline normative expectations (Geist, 2009).

2.3 Underlying Norms on Family Planning

The reunification of Germany in 1990 brought these two diametrically opposed systems under a single federal and legal framework, forcing the sudden integration of populations with vastly different demographic behaviors. Although the fall of the Berlin Wall meant that East Germany's policies would dissipate and the reunified Germany would follow the West German model, the structural reality in terms of childcare availability could not be eradicated. Kreyenfeld and Konietzka (2017) state that even after the reunification, public childcare was still more accessible in the eastern states than in the western ones. In 1990, the percentage of children aged 0-2 who were enrolled in public childcare was 56 percent for East Germany and only 2 percent for West Germany. The percentage for eastern states declined slightly after the reunification, and stabilized itself at around 40 percent, but western states still saw enrollments under 7 percent until 2007.

Moreover, Döge and Keller (2012) use the survey responses of university students to investigate whether young adults' ideas on parenting and early institutional childcare still reflect the diverse socio-cultural conditions and traditions of the formerly divided Germany. They discover that West German young adults still exhibit a lingering cultural reservation towards early institutional childcare, preferring an older starting age and shorter daily durations than their East German peers. Almost a tenth of the West German sample rejected early childcare entirely, compared to near-universal acceptance (98 percent) among East Germans.

Beyond these underlying cultural norms that could lead to divergent fertility intentions between eastern and western women, socio-psychological research identifies three distinct, interacting cognitive components that generally explain the formation of these intentions. As Ajzen and Klobas (2013) explain, the first component is defined as Attitudes. This represents the individual's subjective perception of the positive or negative consequences of having a child. This process involves weighing the psychological and emotional value of children against the anticipated loss of personal freedom, financial stability, or career progression. The aggregate of these specific behavioral beliefs forms a generally favorable or unfavorable attitude toward the specific goal of maternity. The second component is Subjective Norms. This dimension captures the perceived social pressure to conform to the expectations and behaviors of known individuals or society overall. A woman's motivation to comply with these referents significantly influences her intention. The third, and arguably most critical component for policy evaluation, is Perceived Behavioral Control. This reflects the individual's confidence in their ability to successfully manage, facilitate, and afford having and raising a child. It encompasses the perceived access to financial resources, adequate housing, partner support, and state infrastructure (Ajzen and Klobas, 2013).

Finally, fertility intentions are not solely individual psychological constructs, and the research conducted by Morosow and Trappe (2018) reiterates that they are subject to inter-generational transmission and peer contagion. Their paper looks at how family planning ideas are passed down from mothers to daughters. It notes that the East German cultural attitude historically admired working mothers and encouraged very early family formation. In contrast, West Germany historically associated maternal employment with negative impacts on child development. Therefore, they delayed childbearing and prioritized achieving their desired career goals first. The authors find that the inter-generational transmission of these distinct fertility timelines remains visible in the data long after reunification.

With these elements in mind, we can develop a broader understanding of how fertility intentions are formed, and why they might differ between East and West Germany. This, in turn, allows us to now investigate the relationship between childcare accessibility and fertility intentions.

Chapter 3

Data

This section describes the data sources and sample characteristics used to investigate the relationship between childcare accessibility and fertility dynamics. To construct a reliable analysis, this paper integrates **three complementary datasets**: Germany's Regional Database for objective infrastructure measures, the German Socio-Economic Panel for individual pregnancy planning timelines, and the Generations and Gender Survey for short-term fertility desires.

3.1 Germany's Regional Database

Our first data source is Germany's Regional Database (Regionaldatenbank Deutschland) for information on objective measures, such as, for example, the number of daycare facilities for children aged 0-3. This database is the official joint product of the German Federal and State Statistical Offices and it is governed by the Federal Statistics Act (Bundesstatistikgesetz), which mandates scientific independence and neutrality in data collection.

We use this source to evaluate the evolution of the number of daycare facilities for children aged 0-3 between 2008 and 2013. Germany's Regional Database facilitates access to this information at an aggregate level, as well as at a state level. This allows us to visually deduct whether the East-West division is still present today, and confirm our hypothesis on whether or not former West German states required a more intense intervention than their Eastern counterparts. While in some states we see a low to medium increase in the number of daycare facilities, there are 6 states that show very clear patterns.

In Figures 3.1 - 3.3, we observe the evolutions of three East German states: Sachsen, Sachsen-Anhalt and Thüringen. The plots demonstrate insignificant structural changes during the period of interest. Conversely, Figures 3.4 - 3.6 underline that West German states such as Schleswig-Holstein, Bayern and Niedersachsen, were confronted with substantial changes to their environment. This initial analysis allows us to decide which states to further focus on, and further in our causal analysis, identify which states can serve as a "treatment group" and which states can be considered their "control counterparts".

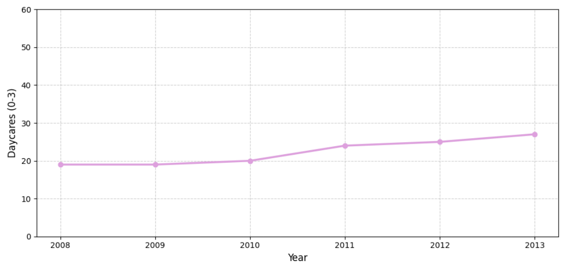

Figure 3.1: The evolution of the number of daycare facilities in Sachsen

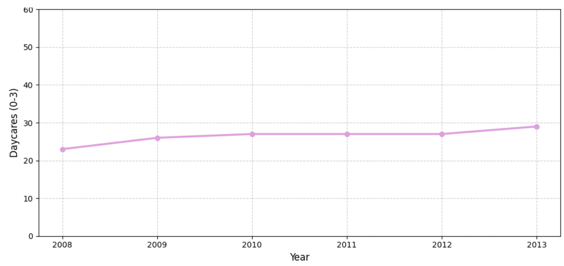

Figure 3.2: The evolution of the number of daycare facilities in Sachsen-Anhalt

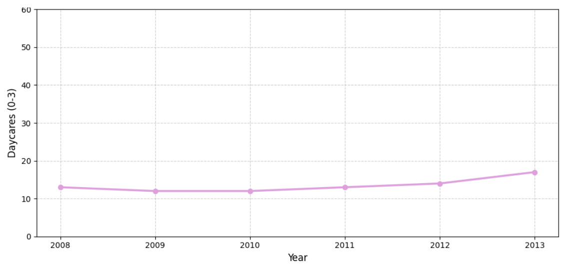

Figure 3.3: The evolution of the number of daycare facilities in Thüringen

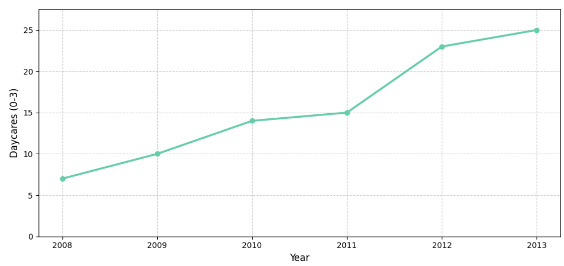

Figure 3.4: The evolution of the number of daycare facilities in Schleswig-Holstein

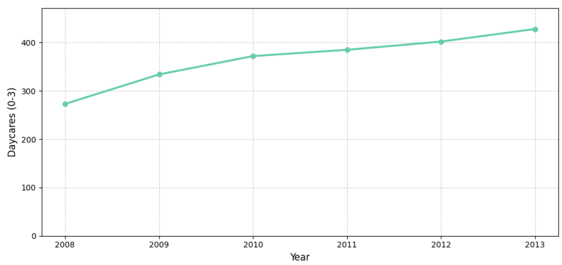

Figure 3.5: The evolution of the number of daycare facilities in Bayern

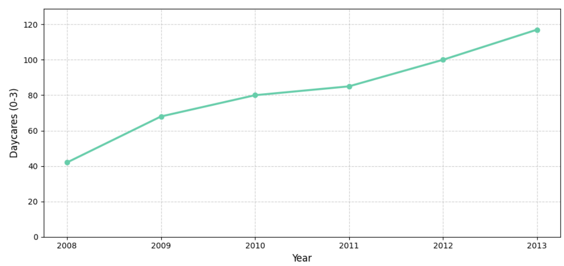

Figure 3.6: The evolution of the number of daycare facilities in Niedersachsen

3.2 German Socio-Economic Panel (SOEP)

Furthermore, we use the German Socio-Economic Panel (SOEP), which is administered by the German Institute for Economic Research (DIW Berlin). The SOEP is a wide-ranging, representative longitudinal study designed to provide high-quality panel data on household dynamics, employment trajectories, and individual attitudes in Germany. To capture the shift in strict fertility timing and evaluate the impact of the 2008 Childcare Funding Act, we exploit panel data from the years immediately preceding and following the reform.

In studying the relationship between **satisfaction with the available childcare** and **strict fertility timing**, we will consider all 16 German states, and we will investigate the variables over a 20 year window, between 2004 and 2024. Overall, we study a sample of German women and whether their last pregnancy was more planned or more unplanned. Table 3.1 provides detailed information on how we use the SOEP to extract and measure the variables relevant for the first part of our analysis.

Table 3.1: Description of Variables and Measurements for SOEP Data

Variable	**Measurement**
The Status of a Pregnancy	Binary indicator: 1 = "more planned"; 0 = "more unplanned".
Employment Status	Binary measure: 1 = full- and part-time workers; 0 = non-working individuals.
Satisfaction with Childcare	Level indicator: Scale from 0 (completely dissatisfied) to 10 (completely satisfied).
Marital Status	Categorical variable: "Unmarried" (baseline/never married), "Married" (including registered partnerships), and "Previously Married" (divorced, annulled, or widowed).
Control Variables	Several continuous demographic variables: the respondent's age in years, the number of children currently living in the household, years of formal education.

This sample is also useful in exploring the third part of our research question. For this, we will focus on individuals residing in Bayern, Niedersachsen, Sachsen, Sachsen-Anhalt, Schleswig-Holstein, and Thüringen, and we will restrict our time frame to the years between 2005 and 2012. Given that Bayern, Niederesachsen and Schleswig-Holstein are West German states, while the rest are East German states, it is crucial to analyze the distribution of demographic characteristics not only within the overall sample, but also across these two distinct regional sub-samples. In our causal analysis, the West German states represent treatment group candidates, while the residents of Sachsen, Sachsen-Anhalt and Thüringen will form a synthetic control group.

Table 3.2 presents the summary statistics for the full sample covering the years from 2004 to 2024, which consists of $N = 10,362$ observations. Examining the baseline demographic characteristics, the individuals in this sample have a mean age of 32 years and report an average of 12.2 years of education. Household sizes reflect standard familial norms, with the mean number of children standing at 2 per household. Regarding family structure and fertility outcomes, a strong majority of the pregnancies observed are planned, accounting for 69 percent ($n = 7,184$) of the total sample. This relatively high rate of planned pregnancies aligns closely with the observed marital stability within the group, as $n = 4,461$ respondents are married. In contrast, $n = 1,267$ remain unmarried, while a small minority of $n = 244$ individuals report being previously married. Turning to the primary independent variable, the data reveals an overall positive perception of the available childcare. The mean childcare satisfaction level is measured at 7.02 on a 10-point scale. Finally, looking at labor market participation, slightly more than half of the sample is actively engaged in the workforce, with employed individuals making up 54 percent ($n = 5,566$) of the total observations.

Table 3.3 presents the summary statistics focusing specifically on the critical policy window from 2005 to 2012. The total sample consists of $N = 1,909$ observations, and we see that the treatment group is substantially larger ($n = 1,299$) than the control group ($n = 610$). Demographic characteristics such as age and education remain relatively consistent across the groups, though slight differences emerge. Individuals in the treatment group are, on average, older (33 years old) than those in the control group (31 years old). Furthermore, both groups show very similar levels of education, with the control group averaging slightly higher at 12.71 years compared to the treatment group's 12.62 years. Significant divergences are observed regarding family structure.

Table 3.2: Summary Statistics: Total Sample, 2004 - 2024

Characteristic	**N = 10,362**
Planned Pregnancies (n, %)	7,184 (69%)
Mean Childcare Satisfaction (0-10)	7.02
Mean Age	32
Mean Number of Children	2
Mean Years of Education	12.2
Employed Individuals (n, %)	5,566 (54%)
Family Status	
Unmarried (n)	1,267
Married (n)	4,461
Previously Married (n)	244

Notes: This table contains summary statistics such as mean, number or percentage, for the data extracted from the German Socio-Economic Panel. It contains information about the number of planned pregnancies, as well as potential determinants, such as the number of employed individuals or the number of married individuals. The data is recorded annually, between 2004 and 2024 and consists of 10,362 observations. *Source:* Own calculations, using data extracted from the German Socio-Economic Panel.

Table 3.3: Summary Statistics: Policy Window, 2005–2012

Characteristic	**Overall** N = 1,909	**Control Group** N = 610	**Treatment Group** N = 1,299
Planned Pregnancies (n, %)	1,356 (71%)	399 (65%)	957 (74%)
Mean Childcare Satisfaction (0-10)	7.24	7.50	7.11
Mean Age	32	31	33
Mean Number of Children	2	2	2
Mean Years of Education	12.65	12.71	12.62
Employed Individuals (n, %)	1,178 (62%)	379 (62%)	799 (62%)
Family Status			
Unmarried (n)	208	129	79
Married (n)	548	134	414
Previously Married (n)	34	15	19

Notes: This table contains summary statistics such as mean, number or percentage, for the data extracted from the German Socio-Economic Panel. The total sample consists of 1,909 observations. We additionally split it between a Control Group (consisting of the East German states: Sachsen, Sachsen-Anhalt and Thüringen) and a Treatment Group (consisting of the West German states: Bayern, Niederesachsen and Schleswig-Holstein). It contains information about the number of planned pregnancies, as well as potential determinants, such as the number of employed individuals or the number of married individuals. The data is recorded annually, between 2005 and 2012. *Source:* Own calculations, using data extracted from the German Socio-Economic Panel.

Marriage rates are much higher in the treatment group, starkly contrasting with the control group. This reflects the FRG's long-standing legacy, and could indicate that the popularized traditional family values still persist today. Alongside these vast differences in family composition, the rate of planned pregnancies also shows a noticeable gap, sitting at 65 percent for the control group and rising to 74 percent for the treatment group. Regarding the remaining variables, the data reveals a nuanced picture of childcare satisfaction and labor market participation. Subjective childcare satisfaction is slightly higher in the control group (mean of 7.50) than in the treatment group (mean of 7.11). However, employment levels show no gap whatsoever, with exactly 62 percent of individuals employed in both the control and treatment groups. Once again, the mean number of children stands at exactly 2 per household across all cohorts. Overall, these statistics provide a foundational understanding of the sample during the policy window.

3.3 Generations and Gender Survey (GGS)

Finally, we use the Generations and Gender Survey (GGS), which is the central data collection instrument of the international Generations and Gender Programme (GGP). The GGP is a multi-disciplinary, cross-national research infrastructure designed to provide high-quality panel data on family dynamics, demographic behaviour, and life-course trajectories. To study the relationship between **satisfaction with the division of childcare responsibilities between spouses** and **short-term fertility desires**, we use Waves 1 and 2 of the first GGS Survey, leveraging data from the years 2005 and 2008. Table 3.4 provides detailed information on the variables we use in this part and how they are measured.

Table 3.4: Description of Variables and Measurements for GGS Data

Variable	**Measurement**
Short-Term Fertility Intentions	Binary indicator: 1 = "Respondent intends to have a child in the next three years"; 0 = "Not".
Satisfaction with the Division of Childcare Labor	Level indicator: Scale from 0 (completely dissatisfied) to 10 (completely satisfied).
Self-Reported Health Status	Level indicator: Scale from 1 (very good health) to 5 (very bad health).
Employment Status	Binary measure: 1 = full- and part-time workers; 0 = non-working individuals.
Use of Institutional Childcare	Binary measure: 1 = "Children are enrolled in a nursery/ kindergarten"; 0 = "Not."
Control Variables	Several continuous demographic variables: the respondent's age in years, the number of biological children, the highest level of completed education.

The summary statistics from Table 3.5 provide an extensive overview of the demographic and socioeconomic landscape for a total sample of $N = 1,634$ individuals across Waves 1 and 2. The participants display a mean age of 36 years and an average of 2 biological children per respondent.

Table 3.5: Summary Statistics: Total Sample, Waves 1 and 2

Characteristic	**N = 1,634**
Short-Term Fertility Intentions (n, %)	272 (16.6%)
Mean Satisfaction with Division of Childcare Tasks (0-10)	8.13
Mean Age	36
Mean Number of Children	2
Mean Completed Levels of Education	3.6
Employed Individuals (n, %)	1,200 (73.4%)
Mean Health Status	1.77
Mean Number of Respondents using Institutional Childcare (n, %)	700 (42.8 %)

Notes: This table contains summary statistics such as mean, number or percentage, for the data extracted from the Generations and Gender Survey. The total sample, consisting of 1,634 observations, is constructed by merging the 2005 Wave with the 2008 Wave. It contains information about the number of women intending to have a child in the following three years, as well as potential determinants, such as the number of employed individuals or the highest level of education completed. *Source:* Own calculations, using data extracted from the Generations and Gender Survey.

The mean level of completed education is 3.6, which is placed between higher secondary education and post-secondary, non-tertiary education. From an economic perspective, the sample exhibits a high degree of labor market participation, as 1,200 individuals are classified as employed in either full-time or part-time positions. Furthermore, the physical well-being of the cohort is notably strong. With health status measured on a scale of 1 to 5, the mean response of 1.77 suggests that most participants view their physical condition favorably. Domestic life also appears highly functional, particularly regarding the sharing of responsibilities. The mean satisfaction score for the division of childcare tasks is 8.13 on a scale ranging from 0 to 10, indicating that the vast majority of individuals are very satisfied with how these duties are distributed within their households. Regarding external support systems, 42.8 percent of the respondents, currently have their children enrolled in institutional childcare facilities like nurseries or kindergartens. However, a striking contrast emerges when examining short-term fertility intentions, defined as the desire to have a child within the subsequent three-year period. Despite the stability in health, employment, and domestic satisfaction, only 16.6 percent of the sample, totaling 272 individuals, intend to have another child in the near future.

Moreover, Wave 1 of the first survey round contains an important question, which helps us better understand the way women form fertility intentions. More specifically, the participants are given a list of 9 factors and for each factor, they are asked to state whether or not it would influence their decision to have another child.

Figure 3.7 illustrates the aggregate preferences of women regarding fertility decisions for the overall sample. The distribution demonstrates that economic and professional considerations dominate the decision-making process. A woman's personal career emerges as the most regarded factor, with approximately 44 percent of respondents strongly considering it, closely followed by her overall financial situation at nearly 39 percent. Crucially, these economic variables substantially outrank relational and biological factors, such as finding a suitable partner (29 percent) or personal health

(20 percent). Furthermore, structural support systems play a pivotal role, with childcare availability (29 percent) and the opportunity for parental leave (27 percent) serving as critical determinants. For the purpose of evaluating the 2008 Childcare Funding Act, this baseline distribution is highly instrumental. By addressing the structural deficit in childcare availability, the 2008 legislation theoretically mitigates the primary economic impediments, facilitating a dual-earner model where women are not forced to choose between fertility intentions and career continuity.

Furthermore, we are interested in analyzing the rank at a regional level, making a split between East and West Germany. Although West Berlin was situated geographically and politically in West Germany, it functioned as a distinct demographic and social anomaly whose characteristics frequently mirrored East German cultural norms more closely than those of the broader Federal Republic of Germany. For example, we know that only 2 percent of the children in West Germany aged under three were enrolled in nurseries. However, nearly 20 percent of the children in this group lived in West Berlin (Ahnert and Lamb, 2001). In the years after the reunification of Germany, the community of West Berlin saw a behavioral and structural convergence towards the landscape of the East. Because of this integration, contemporary demographic analyses investigating fertility rates and intentions methodologically classify the entirety of Berlin as part of East Germany. Klüsener and Goldstein (2016) state that this approach is practically necessitated by data constraints within official birth registries, which often no longer differentiate between the former East and West Berlin. Additionally, Bujard and Huebener (2024) argue that this methodological choice is robust because modern Berlin's geographical position, historical characteristics, and current socio-economic indicators align much more closely with East German characteristics than with those of West Germany. Therefore, this paper also groups the data of Berlin with East Germany.

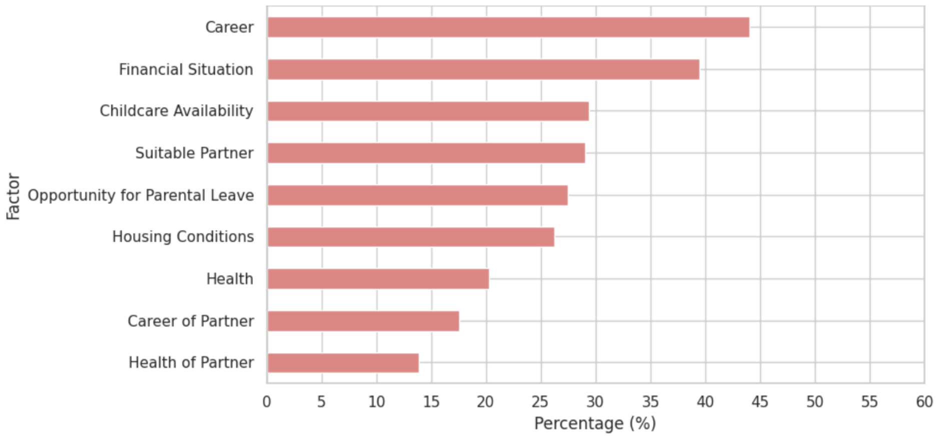

Figure 3.7: General Ranking

Figure 3.8 shows the fertility decision factors exclusively for the East German subsample, revealing a heightened sensitivity to both economic stability and structural support. Career considerations dictate the fertility intentions of approximately 56 percent of respondents, while the financial situation is prioritized by roughly 48 percent. Childcare availability ranks as the definitive third most crucial factor at 35 percent, slightly higher than the opportunity for parental leave (34 percent). This pronounced prioritization can be attributed to the historical legacy of the former GDR, characterized by near-universal female labor force participation and comprehensive state-provided childcare. Consequently, East German women exhibit a culturally embedded expectation of combining motherhood with uninterrupted employment. This structural dependency implies that some exogenous policy shocks, such as the 2008 Childcare Funding Act, will probably be perceived differently based on geographical location.

Finally, Figure 3.9 provides the ranking determined by the responses of West German women. While career (41 percent) and financial situation (37 percent) remain the foremost considerations, their aggregate magnitude is notably diminished compared to the East German cohort. Furthermore, the hierarchy of secondary factors shifts. The necessity of finding a suitable partner is ranked slightly higher (28 percent) than childcare availability (27 percent), indicating a slightly more traditional, relationship-centric approach to family planning. These figures reflect the historical dominance of the male-breadwinner model in the former FRG, where maternal employment was less structurally supported by the state. Consequently, West German women historically faced steeper opportunity costs when attempting to reconcile career aspirations with motherhood due to the systemic lack of childcare infrastructure. Therefore, evaluating the 2008 Childcare Funding Act through a causal framework must acknowledge that the policy introduces a more disruptive structural shift in the West. It fundamentally alters the local constraints on maternal labor supply, potentially driving a more significant behavioral response in female employment and fertility decisions in this region.

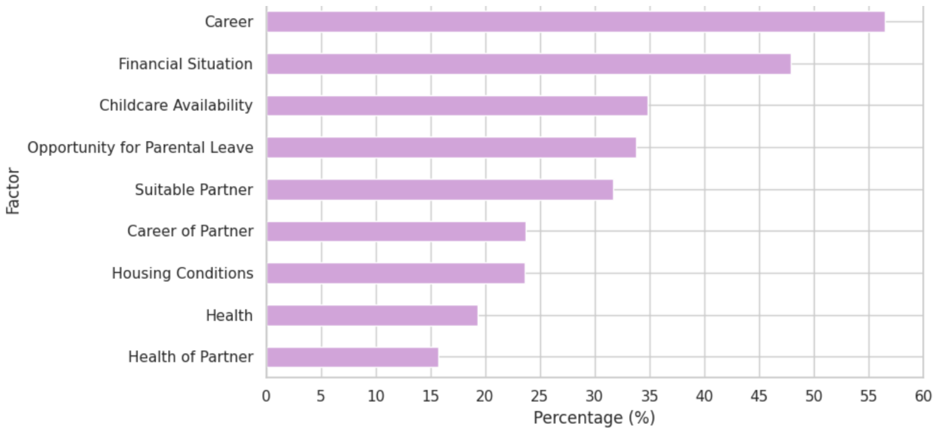

Figure 3.8: Ranking by East German women

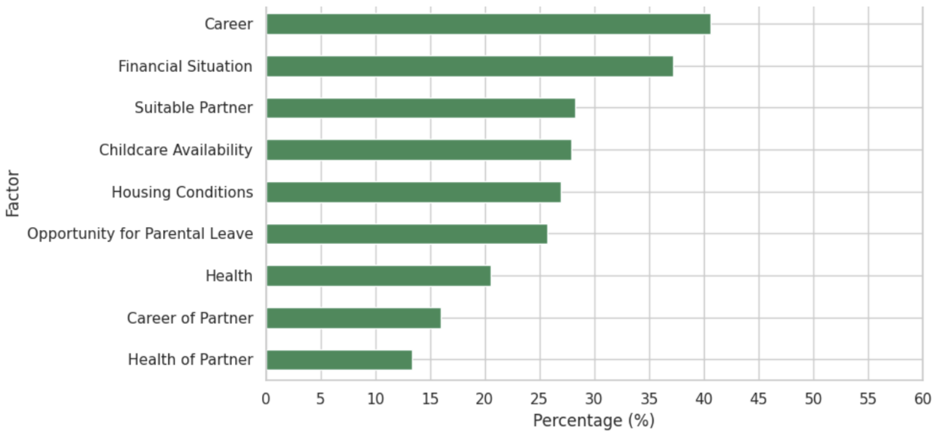

Figure 3.9: Ranking by West German women

Notes: In the 2005 Wave of the Generations and Gender Survey, participating women are given a list of 9 factors and for each factor, they are asked to state whether they strongly take it into consideration when making fertility decisions or not. Thus, for each factor, we can find the percentage of women that consider it relevant when planning to have a child. Figures 3.7 - 3.9 display the rankings obtained through these percentages, at first for the total population of Germany, and then split between East and West Germany. The descriptions state the region to which each ranking corresponds. *Source:* Own calculations, using data from the Generations and Gender Survey 2005 Wave.

Overall, the summary statistics from these combined data sources reveal important baseline insights, but they also reveal a thin slicing of data within our restricted policy window may introduce potential volatility and small-sample biases.

Chapter 4

Empirical Strategy

This section outlines the econometric models used to analyze the relationship between childcare support and family planning. To evaluate correlational patterns, we establish a random-effects binary choice model (using Probit and Logit specifications), which accounts for unobserved individual heterogeneity over time. To isolate causation, we introduce a Synthetic Difference-in-Differences model.

4.1 Random Effects Probit and Logit Models

4.1.1 Family Planning and Pregnancies

In order to analyze the first part of our research question, we employ a binary choice framework, and we are interested in the correlation between strict **fertility timing** and **satisfaction with the available childcare facilities.**

In the previous section, we discussed how the responses were quantified for each variable of interest, and we use this information to construct our model as follows. Given that observations are nested within individuals over time, we employ a Random Effects (RE) framework. This approach allows us to account for unobserved individual-specific heterogeneity, α_i, such as stable personality traits or family-oriented values that do not change over the observed horizon.

We assume the existence of an unobserved latent variable, Y_{it}^*, representing the likelihood of an individual i experiencing a planned pregnancy in period t.

$$Y_{it}^* = \alpha_i + \theta \text{Childcare}_{it} + \gamma' \mathbf{Z}_{it} + \varepsilon_{it}$$

Where:

- Childcare_{it} is the primary explanatory variable, measured as the level of satisfaction with the available childcare.
- $\mathbf{Z}_{it} \in \mathbb{R}^5$ is a vector of socio-economic control variables. Each element represents one of the following variables: age, number of children in the household, years of education, employment status, and family status.
- ε_{it} is the idiosyncratic error term.

The actual observed outcome is binary, and it is determined by the threshold rule:

$$Y_{it} = \begin{cases} 1 & \text{if } Y_{it}^* > 0 \\ 0 & \text{if } Y_{it}^* \leq 0 \end{cases}$$

Let $\mathbf{X}_{it} \in \mathbb{R}^6$ be the vector of information corresponding to respondent i at time t, where the first element is Childcare$_{it}$ and the following five elements are $\mathbf{Z}_{it}$. Additionally, let $\beta \in \mathbb{R}^6$ be the vector of coefficients, where the first element is θ, and the following five elements are γ.

4.1.2 Short-Term Fertility Intentions

For the second part, we once again employ a binary choice framework, and we are interested in the correlation between **short-term fertility desires** and **satisfaction with the division of childcare responsibilities between spouses.** In a similar manner, we previously discussed how the responses were quantified for each variable of interest, and we use this information to construct our model as follows. Once again, we employ a Random Effects (RE) framework.

We assume the existence of an unobserved latent variable, Y_{it}^*, representing the underlying desire of an individual i to have a child in the next three years, measured in period t.

$$Y_{it}^* = \alpha_i + \theta \text{Labor Satisfaction}_{it} + \gamma' \mathbf{Z}_{it} + \varepsilon_{it}$$

Where:

- Labor Satisfaction$_{it}$ is the primary explanatory variable, measured as the level of satisfaction with the way that domestic childcare responsibilities are split between spouses.
- $\mathbf{Z}_{it} \in \mathbb{R}^6$ is a vector of socio-economic control variables. Each element represents one of the following variables: age, number of biological children, level of education, employment status, health status, and a binary indicator for the use of institutional childcare.
- ε_{it} is the idiosyncratic error term.

The actual observed outcome is binary, and it is determined by the threshold rule:

$$Y_{it} = \begin{cases} 1 & \text{if } Y_{it}^* > 0 \\ 0 & \text{if } Y_{it}^* \leq 0 \end{cases}$$

Let $\mathbf{X}_{it} \in \mathbb{R}^7$ be the vector of information corresponding to respondent i at time t, where the first element is Labor Satisfaction$_{it}$ and the following six elements are $\mathbf{Z}_{it}$. Additionally, let $\beta \in \mathbb{R}^7$ be the vector of coefficients, where the first element is θ, and the following six elements are γ.

4.1.3 General framework

Having these two separate frameworks in mind, we can further consider the overall statistical approach. Our model then becomes:

$$Y_{it}^* = \alpha_i + \mathbf{X}_{it}' \beta + \varepsilon_{it}$$

$P(Y_{it} = 1|\mathbf{X}_{it}, \alpha_i)$, is modeled using a cumulative distribution function (CDF), denoted as $F(\alpha_i + \mathbf{X}'_{it}\beta)$. The choice of F determines the link function used for estimation.

The primary analysis utilizes the Probit model, which assumes the error term follows a standard normal distribution. The link function is the inverse normal CDF:

$$\text{probit}(p_{it}) = \Phi^{-1}(p_{it}) = \alpha_i + \mathbf{X}'_{it}\beta$$

The resulting probability is given by:

$$P(Y_{it} = 1|\mathbf{X}_{it}, \alpha_i) = \Phi(\alpha_i + \mathbf{X}'_{it}\beta) = \int_{-\infty}^{\alpha_i + \mathbf{X}'_{it}\beta} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right) dz$$

To ensure the results are not sensitive to the distributional assumptions of the Probit model, we perform a robustness check using a Logit model. This assumes ε follows a standard logistic distribution. The link function becomes:

$$\text{logit}(p_{it}) = \ln\left(\frac{p_{it}}{1 - p_{it}}\right) = \alpha_i + \mathbf{X}'_{it}\beta$$

The inverse link provides the probability:

$$P(Y_{it} = 1|\mathbf{X}_{it}, \alpha_i) = \frac{\exp(\alpha_i + \mathbf{X}'_{it}\beta)}{1 + \exp(\alpha_i + \mathbf{X}'_{it}\beta)}$$

Because the standard normal and the logistic CDF are non-linear, the estimated coefficients do not represent the marginal change in probability. To provide a meaningful interpretation, we calculate and report Average Marginal Effects (AME):

Marginal effect of variable X_{it_k} in the Probit Model:

$$\frac{\partial P(Y_{it} = 1|\mathbf{X}_{it}, \alpha_i)}{\partial X_{it_k}} = \phi(\alpha_i + \mathbf{X}'_{it}\beta) \cdot \beta_k$$

where ϕ is the standard normal density function. Marginal effect of variable X_{itk} in the Logit Model:

$$\frac{\partial P(Y_{it} = 1|\mathbf{X}_{it}, \alpha_i)}{\partial X_{it_k}} = \lambda(\alpha_i + \mathbf{X}'_{it}\beta) \cdot \beta_k$$

where λ is the logistic density function.

For binary variables, the marginal effect is calculated as the discrete change in probability when the variable switches from 0 to 1, holding all other covariates at their observed values and averaging over the sample.

4.2 Synthetic Difference in Differences Model

To evaluate the impact of the 2008 Childcare Funding Act (KiföG) on **strict fertility timing**, this paper employs a Synthetic Difference-in-Differences (SDiD) estimation strategy. The SDiD estimator compares changes in outcomes between a treatment group and a control group before and after a specific intervention. Germany's historical division provides a natural experiment: while West German states historically lacked comprehensive childcare, East German states maintained near-universal childcare infrastructure. Therefore, the 2008 KiföG acts as an exogenous expansion of childcare (treatment) primarily for West Germany, while neighboring East German states serve as the control group.

To demonstrate that a German state can serve as either a treatment or control candidate, we propose a stability analysis using an index-based approach. Since the treatment consists of an expansion of childcare facilities, it is a reasonable choice to measure the evolution of the number of daycare facilities for children aged 0-3 in our years of interest $T_0 \dots T_n$. Let $T_0 \dots T_m$ denote the pre-treatment periods and $T_{m+1} \dots T_n$ denote the post-treatment periods.

We start by normalizing the number of daycare facilities for children aged 0-3 to the baseline year. This allowed us to effectively control for the absolute size of each state, further enabling a direct comparison of growth trajectories. Let $C_{s,t}$ be the number of daycare facilities for children aged 0-3, in state s at time t. The stability score (SS_s) of state s is calculated as the cumulative absolute deviation of the indexed values from the baseline.

$$I_{s,t} = \left(\frac{C_{s,t}}{C_{s,T_{m+1}}} \right) \times 100$$

$$SS_s = \sum_{t=T_{m+2}}^{T_n} |I_{s,t} - 100|$$

In this framework, a lower SS_s indicates a stable or flat trajectory, whereas a higher SS_s denotes volatility, which, in the context of the expansion period, represents aggressive structural growth. We calculate the stability index for each of the 16 German states, and based on these values, we identify whether a German state can be classified as a potential treatment or control state.

Once our desired treated state is selected, we utilize the Synthetic Difference-in-Differences (SDID) framework to optimally weight the available control states, generating a "synthetic" match that precisely mirrors the treated unit's pre-intervention baseline. The methodology in this section is based on the Synthetic Difference-in-Differences framework developed by Arkhangelsky et al. (2021).

Suppose our panel consists of N total states observed throughout our years of interest $T_0 \dots T_n$. Let N_{co} represent the number of potential control states, also known as the "donor pool", and let $N_{tr} = 1$ represent our treated state. As mentioned before, let $T_0 \dots T_m$ denote the pre-treatment periods and $T_{m+1} \dots T_n$ denote the post-treatment periods.

In order to construct a good synthetic match for our chosen treatment group, we assign a vector of weights ω to the control states such that their weighted average rate of planned pregnancies in the years $T_0 \dots T_m$ matches the one of the chosen treatment state. The optimal unit weights, $\hat{\omega}^{sdid}$, are calculated by minimizing the following penalized objective function:

$$(\hat{\omega}_0, \hat{\omega}^{sdid}) = \arg \min_{\omega_0 \in \mathbb{R}, \omega \in \Omega} \sum_{t=T_0}^{T_m} \left(\omega_0 + \sum_{i=1}^{N_{co}} \omega_i Y_{i,t} - Y_{tr,t} \right)^2 + \zeta^2 Q ||\omega||_2^2$$

Where:

- $Y_{i,t}$ is the rate of planned pregnancies for state i at time t.

- ω_0 is an unpenalized intercept that allows the weighted control trend to be parallel to (instead of identical to) the treated trend. This ensures the estimator remains invariant to additive state-level shifts.
- $\Omega = \left\{\omega \in \mathbb{R}_+^{N_{co}} : \sum_{i=1}^{N_{co}} \omega_i = 1\right\}$ restricts the weights to be non-negative and sum to one.
- ζ is a regularization parameter designed to increase the dispersion of the weights, preventing the model from over-fitting to a single control state and ensuring uniqueness.
- Q represents the number of years in the pre-treatment period.

In addition to state weights, SDID isolates the pre-treatment periods that best predict post-treatment behavior. The time weights, $\hat{\lambda}^{sdid}$, balance the pre-exposure time periods with the post-exposure average for the control group:

$$(\hat{\lambda}_0, \hat{\lambda}^{sdid}) = \arg \min_{\lambda_0 \in \mathbb{R}, \lambda \in \Lambda} \sum_{i=1}^{N_{co}} \left(\lambda_0 + \sum_{t=T_0}^{T_m} \lambda_t Y_{i,t} - \frac{1}{K} \sum_{t=T_{m+1}}^{T_n} Y_{i,t} \right)^2$$

Where:

- $\Lambda = \left\{\lambda \in \mathbb{R}_+^{Q} : \sum_{t=T_0}^{T_m} \lambda_t = 1\right\}$ restricts the pre-treatment time weights to be non-negative and sum to one.
- K represents the number of years in the post-treatment period.

In a regular Difference-in-Differences model, we would employ the standard Two-Way Fixed Effects regression framework:

$$Y_{i,t} = \iota + \gamma d_i + \theta m_t + \beta d_i m_t + \varepsilon_{it}$$

Where:

- $Y_{i,t}$ is the rate of planned pregnancies of state i at time t, and ι is the average outcome for the control group in the pre-treatment period, which becomes our fundamental baseline
- d_i is a binary indicator for the treatment group, and γ is the baseline difference between the treatment and control group before the treatment occurred
- m_t is a binary indicator for the post-treatment time period and θ is the natural change in the outcome for the control group over time, also known as the time trend
- $d_i m_t$ is the interaction term indicating treatment exposure, and β is the average treatment effect on the treated
- ε_{it} represents the error term

Here, we have successfully isolated the effect of the treatment, and we identify that: $\widehat{ATT} = \hat{\beta}$. However, in our Synthetic Difference-in-Differences approach, we must transform this regression into a minimization problem. By utilizing a weighted least squares approach, the model places heavier emphasis on the control states and time periods that are most comparable to the treated environment:

$$(\hat{\beta}^{sdid}, \hat{\iota}, \hat{\gamma}, \hat{\theta}) = \arg \min_{\beta, \iota, \gamma, \theta} \left\{ \sum_{i=1}^{N} \sum_{t=T_0}^{T_n} (Y_{i,t} - \iota - \gamma d_i - \theta m_t - d_i m_t \beta)^2 \hat{\omega}_i^{sdid} \hat{\lambda}_t^{sdid} \right\}$$

Where:

- ι is again the overall intercept.
- γd_i represents the state fixed effects.
- θm_t represents the time fixed effects.
- $d_i m_t$ is the interaction term indicating treatment exposure.
- β is the Average Treatment Effect on the Treated (ATT), representing the isolated causal impact of the KiföG expansion.

To evaluate the statistical significance of our SDID estimates, we implement Placebo Variance Estimation, corresponding to Algorithm 4 in the framework developed by Arkhangelsky et al. (2021).

The procedure works by iteratively reassigning the "treatment" status to units within the control group. The SDID method is then applied to calculate a treatment effect for this fake scenario. Because we know the control state did not actually experience any policy change, any estimated effect is purely the result of random noise in the underlying data. By repeating this permutation process 1,000 times, the model generates an empirical distribution of these placebo effects. This distribution provides a clear picture of how much the data naturally varies without any real intervention.

The variance of this placebo distribution serves as our standard error. Using this standard error, we assess statistical significance through two standard metrics:

1. 95% Confidence Interval: We construct the interval defined as $\hat{\beta}^{sdid} \pm 1.96 \times SE$. If this 95% confidence interval does not contain zero, the estimated treatment effect is considered statistically significant at the 5% level.

2. P-value: We calculate a z-score ($\hat{\beta}^{sdid}/SE$) and derive the corresponding p-value using a standard normal approximation. A p-value strictly less than 0.05 allows us to reject the null hypothesis, concluding that the KiföG policy expansion had a statistically significant impact on the rate of planned pregnancies.

The validity of our SDID results is based on the following assumptions:

Assumption 1: Parallel Trends After Weighting: The counterfactual trajectory of the treated state is parallel to the synthetic control state after applying the optimal unit weights ($\hat{\omega}^{sdid}$).

$$E[\widehat{Y_{tr,t}}] = \omega_0 + \sum_{i=1}^{N_{co}} \hat{\omega}_i^{sdid} E[Y_{i,t}] \quad \forall t > T_m$$

Assumption 2: Additive Shift Invariance: The baseline outcomes are permitted to differ between the treatment and control groups. Because of the inclusion of state fixed effects and the intercept in the weighting process, the estimator remains unbiased by constant, unobserved baseline heterogeneity.

Assumption 3: No Anticipation Effects: The treatment group must not alter their behavior prior to the treatment implementation.

Assumption 4: No Contemporaneous Shocks: There must be no other changes or policy interventions occurring exactly when the treatment starts that could influence the outcome.

Assumption 5: Stable Unit Treatment Value Assumption (SUTVA): There must be no spillover effects from the treatment group to the control group and the individuals in the treatment group should receive the same "treatment dose".

Assumption 6: Homoskedasticity Across Units: The error terms are homoskedastic across units.

$$Var(\varepsilon_{i,t}) = \sigma^2 \quad \forall i \in \{1, \ldots, N_{co}\}$$

By combining individual-level random-effects regressions with state-level synthetic control simulations, this dual framework allows us to balance micro-level motivations with macro-level policy shocks. Furthermore, our implementation of placebo variance permutations ensures that the standard errors remain reliable despite our regional constraints.

Chapter 5

Results

This section presents the statistical findings derived from our random-effects regressions and Synthetic Difference-in-Differences models. We report the average marginal effects regarding parental satisfaction with the available childcare facilities and satisfaction with the division of childcare tasks between spouses, before presenting our causal average treatment effect on the treated across our selected West German states.

5.1 Family Planning and Pregnancies

Prior to running the regression models, we generate a correlation matrix to examine the linear relationships between the independent variables. Overall, the results in Figure 5.1 show that most variables have weak to moderate relationships with one another (falling between -0.3 and 0.3), which means the variables are distinct enough to be included together in our models without heavily skewing the results.

Figure 5.1: Correlation Matrix: SOEP Data

The most striking value in the matrix is the strong negative correlation (-0.88) between being married and being unmarried. This result is unsurprising, as these are dummy variables created from the same underlying family status category. Because they are mutually exclusive, they naturally move in opposite directions.

Importantly, our main independent variable of interest shows very little correlation with the demographic controls. Its strongest relationship is a weak positive correlation with the number of children (0.13), while its correlations with age (0), education (-0.03), and employment (-0.04) are nearly non-existent. This lack of strong correlation is highly beneficial for our analysis, as it indicates that childcare satisfaction operates as an independent factor.

In addition to proving the absence of multicollinearity, we must also analyze the independence of observations. Because our dataset utilizes panel data spanning from 2004 to 2024, it inherently includes multiple observations of the same respondents over time, which violates this assumption. To correct for this, we employed a random-effects modeling approach, as described in the Empirical Strategy chapter of this paper. Finally, we must provide an adequate sample size. With $N = 4,046$ observations and $3,235$ individual mothers, our models possess sufficient statistical power to generate highly reliable results.

Table 5.1: Average Marginal Effects using SOEP Data

	Probit Results	**Logit Results**
Childcare Satisfaction	0.012***	0.016***
	(0.00)	(0.00)
Age	0.000	0.000
	(0.00)	(0.00)
Number of Children in HH	-0.075***	-0.071***
	(0.01)	(0.01)
Years of Education	0.020***	0.022***
	(0.00)	(0.00)
Employment Status	0.000	0.000
	(0.02)	(0.02)
Respondent is Married	0.194***	0.191***
	(0.02)	(0.02)
Respondent was Previously Married	-0.026	-0.027
	(0.04)	(0.04)

Notes: This table presents the average marginal effects estimated from both Probit and Logit models. These models estimate the likelihood of an individual experiencing a planned pregnancy. The independent variables capture socio-demographic characteristics and potential determinants, including childcare satisfaction, age, the number of children in the household, years of education, employment status, and marital status. Standard errors are reported in parentheses directly below the marginal estimates. Statistical significance is denoted by *** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$. The models are estimated using a sample of 4,046 observations. *Source:* Own calculations, using data extracted from the German Socio-Economic Panel.

Table 5.1 presents the Average Marginal Effects derived from both Probit and Logit regressions, modeling the likelihood of a planned pregnancy. The results from both the Probit and Logit models are almost identical at the reported decimal level, demonstrating that the findings are robust and not a byproduct of the underlying mathematical assumptions of each model. Because we are looking at Average Marginal Effects, the coefficients can be interpreted as the percentage point change in the probability of a planned pregnancy given a one-unit change in the independent variable.

Starting with the primary variable of interest, childcare satisfaction shows a positive and highly statistically significant relationship with planned pregnancies. Specifically, a one-unit increase in a respondent's childcare satisfaction score is associated with a 1 percentage point increase in the probability of a planned pregnancy. While this effect might appear numerically small at first glance, it is highly precise. In the context of a 10-point satisfaction scale, shifting from the lowest to the highest level of satisfaction could substantially alter a household's family planning dynamics.

The models highlight several important control variables that significantly influence family planning. Firstly, being married is the strongest factor in the model. Married respondents are 19 percentage points more likely to report a planned pregnancy compared to the baseline group, which consists of unmarried individuals. Furthermore, the existing size of the household has a strong deterrent effect on future planned pregnancies. For every additional child currently in the household, the probability of a subsequent planned pregnancy decreases by 7 percentage points. Finally, similarly to the marital status and the satisfaction with the available childcare, the number of years spent in school is positively associated with planned pregnancies. Each additional year of education increases the likelihood of a planned pregnancy by 2 percentage points, suggesting that individuals with longer educational backgrounds may exercise more deliberate control over their fertility timing.

Moreover, it is equally important to acknowledge the variables that do not show a statistically significant effect. On one hand, age has a coefficient of almost 0 and lacks statistical significance. This implies that, once we control for factors like marriage, education, and existing children, a respondent's age does not meaningfully change the statistical probability of their pregnancy being planned. In addition, being divorced or widowed shows a slight negative coefficient, but this result is statistically insignificant. Therefore, we cannot confidently say that previously married individuals differ from unmarried individuals regarding planned pregnancies in this sample. In the end, employment status does not seem to have a significant impact on the likelihood of a planned pregnancy.

Overall, these results successfully demonstrate that higher childcare satisfaction positively influences the likelihood of a planned pregnancy. Furthermore, the outcome is heavily shaped by an individual's educational background, their current family size, and their marital status.

5.2 Short-Term Fertility Intentions

Since we explore the second part of our research question using the same methodology as above, we, once again, begin our analysis with the correlation matrix. One of the more notable findings is the positive correlation between a respondent's age and their number of biological children (0.28). This is a predictable result, as older individuals have had more time to start or expand their families. We also observe a moderate positive relationship between age and education (0.31), which likely reflects the typical progression of educational attainment over the life course.

Crucially, our primary variable of interest demonstrates very weak correlations with the other variables in the dataset. For example, its relationship with employment status is nearly non-existent at 0.06. In addition, the correlation with education is essentially zero at -0.01 and it shows only a slight negative correlation with self-reported health status (-0.10). This lack of strong association between childcare satisfaction and the demographic controls is highly advantageous for the study, as it suggests that labor satisfaction operates as an independent factor.

In addition to proving the absence of multicollinearity, we must also analyze the independence

Figure 5.2: Correlation Matrix: GGS Data

of observations. Our dataset contains multiple observations of the same respondents over time, because it utilizes panel data from two years: 2005 and 2008, effectively violating this assumption. As mentioned before, we employed a random-effects modeling approach to control for this. Finally, we must provide an adequate sample size. With $N = 1,634$ observations and the same number of individuals, our models likely do not possess sufficient statistical power to generate highly reliable results, and we must seriously take into consideration the risk of small sample bias.

Table 5.2 presents the average marginal effects from both Probit and Logit regression models, assessing how various factors relate to short-term fertility intentions. Across both models, the results are remarkably consistent in terms of coefficient size, direction, and statistical significance, indicating that the findings are robust.

Starting with the primary variable of interest, satisfaction with labor division shows an extremely small and statistically insignificant relationship with fertility intentions. The average marginal effect is just 0.003 in the Probit model and 0.004 in the Logit model. Because these estimates completely lack statistical significance, we can't conclude that a respondent's satisfaction with the division of household and childcare tasks plays a meaningful role in their short-term plans to have a child.

Instead, the models highlight three demographic control variables that exert a highly substantial and statistically significant influence on fertility intentions. Firstly, the existing number of children has the strongest deterrent effect in the model. For each additional biological child a respondent already has, the probability of intending to have another child decreases by 11.2 percentage points in the Probit model and 9.8 percentage points in the Logit model. Moreover, a respondent's age is also negatively associated with fertility intentions. Every additional year reduces the likelihood of intending to have a child by roughly 1.9 percentage points in the Probit model and 1.8 percentage points in the Logit model. Conversely, educational attainment is positively linked to fertility intentions. A higher level of completed education increases the probability of intending to expand the family by 3.7 percentage points under the Probit model and 3.5 percentage points under the Logit model.

Table 5.2: Average Marginal Effects using GGS Data

	Probit Results	**Logit Results**
Use of Institutional Childcare	0.005	0.004
	(0.02)	(0.01)
General Health Status	-0.021	-0.022
	(0.01)	(0.02)
Age	-0.019***	-0.018***
	(0.00)	(0.01)
Number of Biological Children	-0.112***	-0.098***
	(0.01)	(0.02)
Level of Education	0.037***	0.035***
	(0.02)	(0.02)
Employment Status	-0.004	-0.001
	(0.02)	(0.02)
Satisfaction with Division of Childcare Tasks	0.003	0.004
	(0.03)	(0.04)

Notes: This table presents the average marginal effects estimated from both Probit and Logit models. These models estimate the likelihood of an individual intending to have a child in the next three years. The independent variables capture socio-demographic characteristics and potential determinants, including the use of institutional childcare, general health status, age, number of biological children, level of education, employment status, and satisfaction with division of childcare tasks. Standard errors are reported in parentheses directly below the marginal estimates. Statistical significance is denoted by *** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$. The models are estimated using a sample of 1,634 observations. *Source:* Own calculations, using data extracted from the Generations and Gender Survey.

It is equally important to acknowledge the other variables that fail to show a statistically significant effect. Whether or not a household utilizes institutional childcare shows a negligible positive coefficient and is not statistically meaningful. In addition, the coefficient for health status is negative, which weakly hints that poorer health might discourage childbearing intentions. However, this effect is statistically insignificant. Finally, employment status does not appear to have any noticeable or reliable impact on a respondent's short-term fertility intentions within this sample.

Overall, these results demonstrate that short-term fertility intentions are not heavily driven by a respondent's satisfaction with task division, their employment status, or their use of childcare services. Instead, the outcome is heavily shaped by foundational demographic characteristics: an individual's current family size, their age, and their educational background. Building upon these foundational regression findings, the next section of this thesis will pivot to a causal analysis.

5.3 State-Subsidized Childcare and Strict Fertility Timing

In the third part of our analysis, we employ a Synthetic Difference in Differences (SDID) Model, where we choose the residents of the West German states Schleswig-Holstein, Niedersachsen and Bayern as potential treatment groups, and we pool together the residents of the East German states Sachsen, Sachsen-Anhalt and Thüringen in order to create our control group.

To identify whether a German state is a control or treatment candidate, we conducted the stability analysis explained in the Methodology chapter. Using data from Germany's Regional Database, we analyze the evolution of the number of daycare facilities for children aged 0-3 between 2008 and 2013.

The states that obtained the lowest scores are Sachsen, Sachsen-Anhalt and Thüringen, while the states with the highest scores were Schleswig-Holstein, Bayern and Niedersachsen. We select the West German states as our target treatment states, as we have concluded that their residents have faced the largest structural shock in terms of the availability of childcare, and we include Sachsen, Sachsen-Anhalt and Thüringen in our pool of control states, also known as the "donor pool". By observing these control states, we can estimate what the rate of planned pregnancies would have looked like in our target West German states had the childcare expansion not occurred.

Regarding potential contemporaneous shocks, we choose a restricted time-frame, considering the years 2005-2008 as our pre-treatment period, and the years 2009-2012 as our post-treatment period. Additionally, we believe that any policy that might have taken effect during this period that is related to fertility intentions and family planning would have affected the treatment and control groups in a similar manner. Therefore, potential shocks coming from this would be absorbed by the overall time trend. To address the potential anticipation effects, we studied how the percentage of planned pregnancies has evolved in each of the six German states, in the years before the policy. We did not observe any shifts in behavior that could have indicated that women were already reacting to the news of the policy, before it was actually implemented.

Finally, to satisfy the Stable Unit Treatment Value Assumption, we restrict our sample to only individuals who have not changed their residency throughout the selected time frame. We can't state with precision that all individuals in a treated state receive the same "treatment dose", because the SOEP does not provide detailed information on the respondents' residential area, and the German Regional Database does not offer details on the new facilities' addresses. However, taking into account that the expansion in all three target states was significantly large, we consider this assumption to hold.

Table 5.3 presents the expanded results of our SDID analysis, which aims to isolate the effect of formal childcare expansion on planned pregnancies. Across all three treated states, the model yields negative and highly statistically significant Average Treatment Effect on the Treated (ATT) estimates. Because we utilize a binary outcome variable, these estimates indicate a substantial percentage point decrease in the probability of a planned pregnancy relative to each state's synthetic counterfactual.

Schleswig-Holstein experienced the sharpest decline, with the policy expansion resulting in a 22.14 percentage point decrease. We can be confident in this negative impact, as the 95% Confidence Interval ranges from -0.243 to -0.199 and does not include zero. Bayern exhibited the smallest, yet still highly significant, reduction, with a 9.59 percentage point decrease in planned pregnancies (CI: -0.118 to -0.073). Niedersachsen showed a similarly strong, negative response, demonstrating a 12.65 percentage point drop (CI: -0.148 to -0.104).

When interpreting the results of the SDID model, it is important to clearly understand what the estimated counterfactual actually represents. The drop we observe in planned pregnancies does not mean that an individual woman in our treated state is being directly compared to a woman living in an East German control state. Instead, the East German data is used purely to build a mathematical simulation of the treated state itself. Therefore, this counterfactual shows exactly what the trend of planned pregnancies would have looked like in Schleswig-Holstein if the formal childcare expansion never happened. Because of this, the ATT suggests that, on average across our sample, the probability of a planned pregnancy for a woman in Schleswig-Holstein after the policy was introduced was 22.14 percentage points lower than it would have been in a simulated scenario where no policy was implemented.

Additionally, Tables 5.4 and 5.5 explain how the SDID algorithm constructed the counterfactuals using state and time weights. For instance, Schleswig-Holstein's synthetic control relies heavily on Sachsen (55.5 percent), alongside Thüringen (32.4 percent) and Sachsen-Anhalt (12.1 percent). In contrast, the counterfactuals for Bayern and Niedersachsen distribute the weights much more evenly across the three East German states, with each contributing roughly 30 percent to 37 percent to the synthetic baseline. To effectively align the pre-treatment time trends with the post-treatment period, the model assigned specific weights to the observed pre-treatment years. As expected, the algorithm assigned the exact same time weights across all three target states. The baseline heavily emphasizes the year 2008 (roughly 66.8 percent) and 2006 (roughly 33.2 percent), while completely discarding the years 2005 and 2007.

Table 5.3: SDID ATT Estimates

	ATT Estimate	Standard Error	95 percent Confidence Interval
Schleswig-Holstein	-0.2214***	0.0113	[-0.243 , -0.199]
Bayern	-0.0959***	0.0113	[-0.118 , -0.073]
Niedersachsen	-0.1265***	0.0113	[-0.148 , -0.104]

Table 5.4: Control States' Weights

	Sachsen	Sachsen-Anhalt	Thüringen
Schleswig-Holstein	0.555	0.121	0.324
Bayern	0.376	0.296	0.327
Niedersachsen	0.353	0.335	0.313

Table 5.5: Pre-Treatment Years' Weights

	2005	2006	2007	2008
Schleswig-Holstein	0.0000000	0.3315876	0.0000000	0.6684125
Bayern	0.0000000	0.3315876	0.0000000	0.6684125
Niedersachsen	0.0000000	0.3315876	0.0000000	0.6684125

Notes: These tables present the results and underlying weighting vectors from the Synthetic Difference-in-Differences estimation. The first table reports the Average Treatment Effect on the Treated for three selected West German states (Schleswig-Holstein, Bayern, and Niedersachsen), alongside standard errors and 95 percent confidence intervals. The ATT Estimate in Table 5.3 measures the effect of the 2008 expansion of childcare facilities on the rate of planned pregnancies in the three target western states. Statistical significance is denoted by *** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$. The second table displays the optimal unit weights assigned to the East German donor pool states (Sachsen, Sachsen-Anhalt, and Thüringen) to construct the synthetic counterfactual for each treated state. Using these three eastern states, we are able to construct imaginary copies of the three western states, in the unrealized event in which they didn't receive the childcare expansion. The third table details the pre-treatment time weights applied to the years 2005 through 2008 to balance the pre-exposure trends with the post-exposure average. The weights are equal across all three treated states, because the minimization algorithm which determines these values only depends on the data from the eastern states. *Source:* Own calculations, using data from the German Socio-Economic Panel.

Overall, this SDID analysis reveals a highly significant, consistent, and negative relationship between childcare expansion and planned pregnancies across multiple West German states. While the magnitude of the drop varies, this approach strongly suggests that the structural expansion of formal care facilities in West Germany was followed by a notable suppression of planned pregnancies compared to the East German trend.

Chapter 6

Discussion

As seen in the previous chapter, the overall results present a complex, seemingly contradictory narrative. This section is meant to further investigate the meaning of the provided results.

With regards to the first part of our analysis, while childcare satisfaction is a highly subjective metric, we argue that it serves as a robust composite measure for overall childcare availability. A parent's reported satisfaction inherently captures various dimensions of care, including geographic proximity to local facilities. Therefore, it is highly probable that the positive effect of satisfaction observed in this model already internalizes the benefits of having a facility close to home. Moreover, Schober and Spiess (2015) emphasize the importance of studying satisfaction with the available childcare, rather than the objective availability itself. They find that for mothers of toddlers living in East Germany, satisfaction with local childcare significantly influenced employment decisions. Specifically, mothers living in districts with smaller day care group sizes were more likely to be employed and to increase their working hours. In contrast, for mothers of children under 3 in West Germany, satisfaction with local day cares showed no significant relationship with maternal employment. The authors attribute this to the severe shortage of childcare slots for toddlers in this region.

The paper by Baizán published in 2009 serves as a relevant comparison to our results for this part. The main independent variable is childcare availability, which, as mentioned above, we consider to be a part of overall childcare satisfaction. The study's most prominent finding is that regional day care availability has a significant and positive effect on both first and higher-order births. Similar to our results, Baizán found that the number of years of education has a positive impact on fertility behavior. The author suggests this may be because highly educated women have better possibilities to combine paid work with motherhood and face higher opportunity costs if they prolong interruptions in their careers. We identify a similar result in our analysis, concluding that highly educated women are more likely to have second or higher-order births, or to experience a planned pregnancy.

Secondly, we discuss the lack of statistical significance in our results for the second part. Yoon discovered in 2017 that a supportive environment has a stronger positive effect on actual births than on fertility intentions. Specifically focusing on the spouse's participation, the paper finds that a husband's help with housework and childcare significantly increases the likelihood of having a second child. Even so, only the husbands who spent most time on childcare tasks determined a statistically significant result. These spouses were situated in the highest quartile, meaning they spent at least three hours a day engaging in childcare tasks, and those with lesser amounts of paternal involvement did not yield a statistically significant result. Because minor contributions seem to not have an effect on actual births or fertility intentions, it is a possibility that this is the driver of our results.

In addition, including the use of institutional childcare as a determinant of short term fertility intentions is partially supported by the available literature. A study by Wood (2025) indicates that working mothers who started using formal childcare for their first child have about a 5 percentage point advantage in the overall likelihood of having a second child compared to their matched counterparts who did not use formal childcare. Reasons for our lack of statistical significance could be the small sample of 1,634 observations, or potentially the fact that the surveyed women indicated fertility desires, and we do not measure actual realized fertility.

Moreover, Riederer et al. (2019) support our choice of studying satisfaction with the division of childcare tasks, as this is a stronger driver of fertility intentions than the actual, objective amount of labor. Childbearing intentions are generally higher among respondents who are content with their task division, regardless of whether that division is modern or traditional. Conversely, women who are dissatisfied with a traditional division of chores have the lowest short-term fertility intentions. However, while a modernized division of chores and the satisfaction surrounding it are critical for forming fertility intentions, the division of household work has very little impact on whether couples actually realize those intentions. The division of labor acts as a background factor for planning, but not necessarily as a strict enabler or barrier to the actual birth.

Finally, we explore the underlying mechanisms driving the SDID results. A robust body of literature highlights that the transition into motherhood introduces a profound conflict between career advancement and family responsibilities, through what is defined as the "motherhood penalty". Labor market research by Budig and England (2001) identifies substantial wage penalties for mothers, while Miller (2011) demonstrates that delaying motherhood provides a strong financial benefit, significantly increasing career earnings and wage rates. Because of these severe career and financial setbacks, many women have altered their reproductive behavior by postponing parenthood to build a professional and financial buffer. Consequently, an unintended or early pregnancy shatters this strategy, exposing women to accelerated wage penalties and economic vulnerability, as emphasized by Sonfield et al. (2013).

Therefore, the notable decrease in strictly "planned" pregnancies following the childcare expansion in West Germany does not suggest a reduced desire for children. Rather, it reflects a shift in family planning behavior facilitated by a supportive structural environment. Without institutional support, a high rate of planned pregnancies might signal a highly stressful environment where women feel compelled to meticulously calculate the timing of motherhood simply to protect their careers and financial stability. A reliable state-subsidized childcare system acts as a vital structural buffer. It absorbs domestic responsibilities, providing women with a sense of security that eases the immense pressure to perfectly time a pregnancy. This aligns with findings from Bauernschuster et al. (2013) and Wang et al. (2026), who noted that public childcare expansion reduces work-family conflict, supports career aspirations, and directly generates higher birth rates.

Moreover, the Institutional Background of this paper details on the division in childcare availability and attitudes regarding the use of these facilities between the former East and West German states. Our analysis using Germany's Regional Database also shows that eastern states such as Sachsen, Sachsen-Anhalt and Thüringen make for good control candidates, while western states such as Schleswig-Holstein, Bayern and Niedersachsen have faced serious changes in local availability. In addition to this, the paper by Mätzke (2019) also supports that the former East-West division is still persistent today. By 2015, about 52 percent of children under three in East Germany were in day care, compared to only about 28 percent in West Germany. Furthermore, the type of care differs heavily: in the East, over 76 percent of these day care spots are full-time, while in the West, less than half are full-time

We further discuss the potential issues with data collection on planned and unplanned pregnancies. The academic literature on this subject is still in its early phases, and available papers mostly discuss the importance of including ambivalent answers in our analyses and the impact of survey design and wording. For example, McQuillan et al. (2011) identified a significant group of women who occupy a "grey area" of reproductive intent. Their research found that 23 percent of women described their feelings toward pregnancy as being "okay either way". This group is distinct from those who are actively trying to conceive or actively trying to avoid pregnancy. Women in this ambivalent category often place a different level of importance on motherhood and have unique attitudes toward career success and relationship satisfaction. Therefore, it is important to consider this third category as well, instead of only defining pregnancies as planned or unplanned, desired or undesired, especially since these women make up almost a quarter of the targeted group.

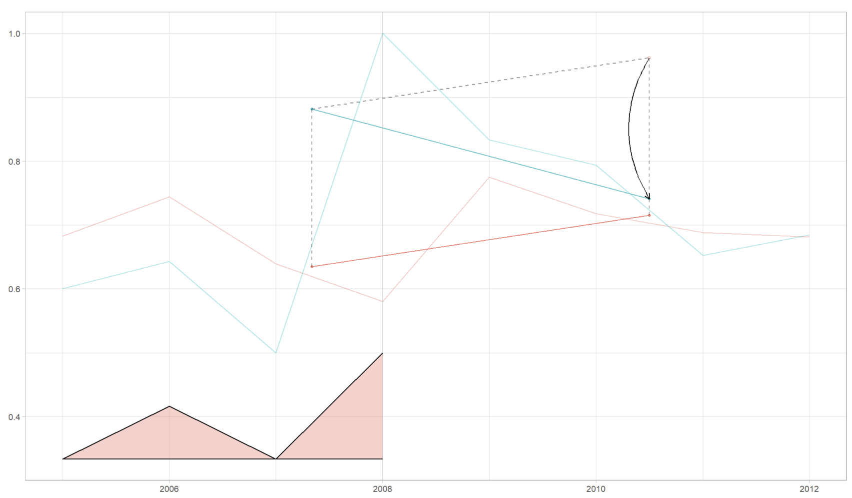

Figure 6.1: Actual vs. Synthetic Trajectories of Planned Pregnancies in Schleswig-Holstein

In addition, Figure 6.1 visualizes the underlying mechanics of the SDID estimation. The blue line represents the actual observed trajectory of planned pregnancies in Schleswig-Holstein, while the red line represents the synthetic control. The red shaded area at the bottom of the graph illustrates the time weights assigned by the algorithm, showing heavy reliance on the years 2006 and 2008 to build the pre-treatment baseline. The dashed line projects the counterfactual post-treatment scenario, and the black arrow represents the final estimated ATT. While the mathematical output of the SDID model indicates a statistically significant negative effect, a visual inspection of this plot suggests we should be careful with the causal interpretation. The most critical vulnerability visible in the graph occurs in 2008, the year immediately preceding the policy expansion. The actual rate of planned pregnancies in Schleswig-Holstein experiences a massive spike, reaching 100 percent. In contrast, the synthetic control remains relatively stable at approximately 65 percent. Because none of the states in the donor pool experienced a comparable spike in 2008, the SDID algorithm mathematically could not construct a synthetic state capable of matching this extreme value. This suggests that the estimated ATT of -22.14 percentage points is biased, as the resulting gap between the actual post-treatment data and the counterfactual is artificially widened.

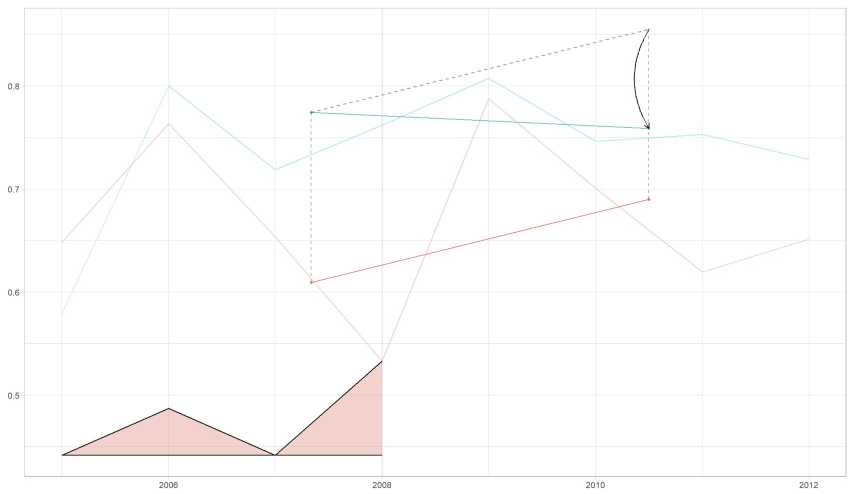

Figure 6.2: Actual vs. Synthetic Trajectories of Planned Pregnancies in Bayern

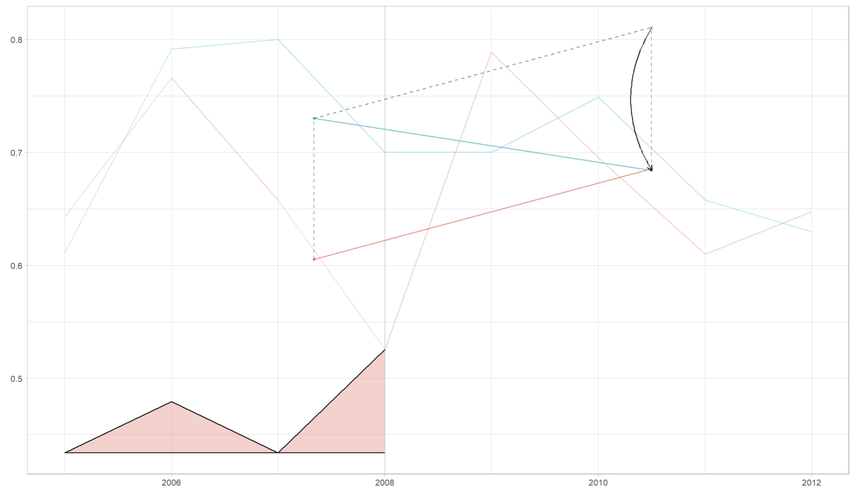

Figure 6.3: Actual vs. Synthetic Trajectories of Planned Pregnancies in Niedersachsen

Notes: These figures visualize the underlying mechanics of the Synthetic Difference-in-Differences (SDID) estimation for the three treated states. In each panel, the solid blue line represents the actual observed trajectory of planned pregnancies, while the solid red line represents the constructed synthetic control. The red shaded area at the bottom of the graphs illustrates the pre-treatment time weights assigned by the algorithm, showing heavy reliance on the years 2006 and 2008 to build the pre-treatment baseline. The dashed line projects the counterfactual post-treatment scenario, and the black vertical arrow highlights the gap between the actual and counterfactual outcomes, representing the final estimated Average Treatment Effect on the Treated. *Source:* Own calculations, using data from the German Socio-Economic Panel.

Similarly, Figure 6.2 illustrates the underlying mechanics of the Synthetic Difference-in-Differences estimation for Bayern. The mathematical output from the SDID model indicates a statistically significant, negative ATT estimate of -9.59 percentage points. A visual inspection of this plot supports a much stronger causal interpretation than in the Schleswig-Holstein case. Pre-treatment parallel trends between actual Bayern and the synthetic control appear relatively robust. While the fit isn't perfect across all years, the fundamental vulnerabilities identified for the previous state do not manifest in Bayern, providing evidence that the estimated ATT reflects a genuine negative relationship rather than being driven by a failure to establish a valid baseline.

Figure 6.3 visualizes the SDID estimation mechanics for Niedersachsen. The model output indicates a statistically significant negative effect, estimating an ATT of -12.65 percentage points. However, visual inspection once again demands caution when interpreting this result as purely causal. A critical vulnerability is clearly visible in 2007. The actual trajectory stagnates at roughly 80 percent, while the synthetic control diverts to roughly 60 percent. Thus, the subsequent counterfactual projection is constructed from an unstable foundation, and we can't exclude the possibility that this artificially inflates the resulting ATT estimate.

This inability to generate a clean pre-treatment fit is potentially rooted in small sample bias. Because we are interested only in pregnant women in specifically six German states, our sample is reduced to only 1909 observations, spread across 8 years. This thin slicing of the data causes extreme volatility, transforming natural, random fluctuations in the survey sample into what appears to be massive structural shocks. Furthermore, this small sample issue extends to the inference strategy. The algorithm is restricted to a donor pool of only three control states. Because the Placebo Variance Estimation relies on iteratively shuffling the treatment status among the control units to calculate the standard error, relying on a donor pool made out of only three states limits the model's ability to accurately capture the true underlying variance.

In summary, our results present a complex narrative regarding fertility behavior. We argue that satisfaction with the available childcare and the division of labor are important drivers of fertility intentions, being complementary to objective metrics. We also hypothesize that the decrease in planned pregnancies following the childcare expansion likely reflects a supportive setting, easing the pressure to rigidly schedule childbirth. However, we must be careful with the causal interpretation of our SDID estimation, as visual inspection reveals critical vulnerabilities in establishing a valid baseline for several states.

Chapter 7

Conclusion

To conclude, we summarize our findings on the complex relationship between childcare infrastructure, household dynamics, and family planning. Using SOEP data, our models initially demonstrated that higher childcare satisfaction positively correlates with planned pregnancies. Simultaneously, GGS data showed that satisfaction with the division of household childcare tasks does not heavily drive short-term fertility intentions. Instead, these are primarily shaped by demographics like age, education, and existing children. However, the SDID framework revealed a contradictory narrative: evaluating the 2008 Childcare Funding Act's expansion in West Germany showed a statistically significant decrease in planned pregnancies compared to synthetic counterfactuals.

Building upon these insights, the results of this study enhance the need for future research. Based on our causal models, we call for additional research to thoroughly investigate the link between state-subsidized childcare and the rate of planned pregnancies, as well as the specific mechanisms through which such policies ease the burden of the motherhood penalty on women. While KiföG expanded legal rights to early childhood support independent of parental employment status, further investigation is needed to see if this, alongside interventions like flexible parental leave or specialized family allowances, universally mitigates the career disruptions women face. Furthermore, executing this future research requires a shift in our approach to data collection. We must ask women more detailed questions about how they form fertility intentions, how they actually navigate the process of planning pregnancies, and which specific factors strongly influence their decisions.

Ultimately, we underline that only with this reliable data and the implementation of detailed surveys can we truly empower women. The global demographic paradigm is shifting, characterized by fertility rates declining to historical lows, largely due to the hardships and demanding pace of contemporary life that make individuals increasingly reluctant to transition into parenthood. Only by talking to women in depth about their real struggles and valuing their opinions can we actually solve the fertility crisis. We should actively address the true impediments and obstacles that women face and that genuinely bother them in order to provide sound, effective policy advice. By thoroughly understanding the impact that structural support systems have on reducing the stress of family planning, policymakers can design interventions that not only sustain macroeconomic labor forces but also genuinely protect a woman's right to safely and freely manage her transition into motherhood.

Chapter 8

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